# Chapter 11
# Manchester as a Digital Powerhouse: Governing the ICT-Related Developments

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**Abstract** The drive to establish Manchester as a forerunner in the digital economy means the city is now considered the "powerhouse" of the North of England, with salaries for city workers at the highest in the UK outside of London. Despite this, Manchester remains home to some of the most socially and economically deprived communities in the UK: just as those who work in the city earn the highest salaries outside London, those who live in the city earn the lowest salaries of the UK's Core Cities. Responding to the particular challenge widespread deprivation poses for a city experiencing economic growth, the administration has deployed an entrepreneurial model of urban development, characterized by close partnerships with private organizations and local actors. This chapter explores the governance of the ICT-related developments responsible for transforming Manchester into a "digital powerhouse" and challenges the city's recently announced "Next Generation Digital Strategy" poses.

## 11.1 Introduction

This chapter is drawn from the case study findings of the city governance project funded by the European Commission's (EC's) Institute for Prospective Technological Studies (IPTS) and EUROCITIES.

Along with Barcelona, Berlin, and Tallinn, Manchester is one of the four emergent city governance systems IPTS and EUROCITES have sought to explore as a

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set of ICT-related developments (Misuraca and Esteve 2010; Misuraca et al. 2011). Setting out Manchester's socioeconomic structure, this chapter outlines the governance system within the city and addresses the challenges these ICT-related developments pose for their emerging digitally inclusive regeneration strategy (Carter 2009; Deakin 2007, 2009). Having done this, the chapter captures the views and opinions of those with a stake in such ICT-related developments and who are affected by the emerging system of governance within the City of Manchester. These insights are drawn from a series of semi-structured interviews, designed by IPTS and conducted by Edinburgh Napier University, with the City of Manchester's academic, industrial, and administrative communities. The interviews in question were undertaken between December 2010 and January 2011.

From here the chapter reflects upon how Manchester proposes to draw upon this governance system as a means to establish itself as a "Next Generation Digital City" (Mycio and Carter 2008). After reflecting on the impacts this latest phase of ICT-related developments are having upon the City of Manchester, the chapter goes on to set out the key findings of this investigation.

## 11.2 The City of Manchester

The City of Manchester covers approximately 117 $km^2$; the population density is over seven times the average for the region and almost ten times the national average. Migration in and out of the city is significant and the city is committed to increasing the number of residents. The population has been increasing at twice the average rate of growth for England and Wales between 2001 and 2007 and at a slightly slower rate in recent years (Manchester Partnership 2007). The 2008-based subnational population projections show that the population of the city will continue to rise to an estimated 519,000 residents by 2015 and 561,800 by 2025 (Manchester Partnership 2009).

Many of Manchester's residents experience significant social and economic deprivation. In 2007, the city was ranked the fourth most deprived local authority in England (Manchester Partnership 2008). Sixty percent of the city's population (234,000) lives in the 10 % most deprived communities in England. Socioeconomic deprivation is described as "widespread across the city" (Manchester Partnership 2009). In 2004, the year the previous English Index of Deprivation (EID) was published, the estimated number of residents living in the most deprived 1 % was 78,599, almost one in five. There has been an improvement in relative terms since 2004, with 2007 figures detailing one in ten Manchester residents living within the 1 % most deprived, approximately 51,155 people (Talukder and Frost 2007).

In terms of the city's geography, the most deprived areas are to the north and the east of the city, but similar patterns of deprivation can be identified using other datasets, such as out-of-work benefit claimants (Manchester Partnership 2008). It is worth noting that the city center continues to improve in terms of its position within

the EID ranking: on the relative scale from 1 to 32,482,[1] the city center moved 7,911 positions between 2004 and 2007, making it the area showing greatest (relative) improvement within the city.

## 11.3 Governance System

Manchester is represented by three tiers of government: Manchester City Council ("local"), UK Parliament ("national"), and European Parliament ("Europe"). The Greater Manchester County Council administration was abolished in 1986 so the city council is, effectively, a unitary authority. Since its inception in 1995, Manchester has been a member of the English Core Cities Group which, among other things, serves to promote the social, cultural, and economic status of the city at an international level.

Under the leadership of the mayor, one third of Manchester City Council is elected each year for three consecutive years, followed by a fourth year without any elections. The Labour Party has had political control in Manchester since 1973. To meet the challenges which the city's commitments to social inclusion and economic competitiveness pose for Manchester, the administration has deployed an entrepreneurial model of urban development, characterized by close partnerships with private organizations and local actors. These partnerships promote interagency working arrangements and funding agreements key to their success. Nowhere is this model better exemplified than in the ICT-related developments the City of Manchester has instigated to underpin their urban regeneration strategy and support this program's social, environmental, and economic aspirations. In terms of their strict chronology, these ICT-related developments can be subdivided into three phases. In line with their underlying technologies, these approximate to online and web-based infrastructures and service platforms supported by next-generation broadband developments.

### *11.3.1 Phase 1: Online Services*

In the first phase (1989–1993), Manchester sought to emulate Germany, Holland, and several Scandinavian countries in their use of ICTs as a means to underpin policy developments and support urban regeneration. This resulted in two initiatives: Manchester Host as the first representation of Manchester as a virtual city and Electronic Village Halls (EVHs) (Digital Cities 2010). These initiatives were aimed

[1] The geographical unit of measurement in the English Indices of Deprivation is the Lower Super Output Area (LSOA); a unit spanning from 1,000 to 1,500 households. There are 32,482 LSOAs in England; therefore, relative deprivation is ranked on a scale from 1 (the most deprived) to 32,482 (the least deprived).

at offering the citizens of Manchester free access to a platform of city services. Entrepreneurial in nature and driven by a consortia of IT consultancies and media companies, Manchester Host did not prove successful. In the first instance, the city's infrastructure was seen as being too heavily dependent on short-term finance and not inclusive enough in terms of the communities it served. While the infrastructure was seen to support the citizens of Manchester, as a service platform, Host failed to include the city's business sector.

### *11.3.2 Phase 2: Web Services*

The second phase, from 1994 to 2002, saw the launch of Virtual Manchester, Manchester online, Virtual Town Hall (VTH), MyManchester, MAD for IT, and Manchester Virtual City. These developments did much to be more inclusive and exploit the opportunities the Internet offered to deliver online services. In reviewing the successes of these developments, Fourkas (2002) singles out the infrastructure underlying the VTH initiative and singles this platform out for particular praise, noting that

> … we should acknowledge the advances [this particular initiative makes] on the delivery of community services through an increasing number of relevant websites and interactive services such as online application forms, email communication with officers, online information, advice, services etc. The 'community' thematic category comes second in online presence rate. (pp. 48–49)

Furthermore, the VTH initiative attracted attention for other reasons: it also provided the infrastructure underpinning the development of Eastserve, a social enterprise project. Local community organizations and third-sector businesses collaborated to recycle and distribute 350 PCs across ten free Internet access points within New East Manchester. The Wired Up Communities initiative went on to distribute a further 3,500 computers. These new or recycled PCs were available at a cost of either £200 or £50, respectively: installation, delivery, and 3 months' dial-up connection were included free, along with 3 h of introductory training. The local credit union provided immediate low-cost loans, and a total of nine local ICT Learning Centers were set up together with a low-cost computer repair service.

Since 25 % of local residents did not have a landline telephone, a wireless broadband network was developed to offer affordable broadband access from £6 per month, without any contract or hidden costs. During this phase, network coverage more than doubled and a further 1,700 PC packages were supplied. Support for the disabled and elderly was added into the residents' service bundles, while wireless networks linked homes, schools, and ICT Learning Centers. With over 5,500 PC packages distributed and over 2,000 regular broadband customers by December 2006, Eastserve had developed into the largest wireless network in Europe.

Reflecting upon the first two phases of ICT development in Manchester, Fourkas (2002) suggests that any future governance strategy in the City of Manchester should:

- Promote such developments under the widest possible partnership between local agencies directly associated with Manchester's urban regeneration strategy.
- Establish a special office that will be in charge of Manchester's Virtual City development, with this agency acting as a coordinating body for the various partners.
- Ensure any developments bridge the digital divide in Manchester by: (a) advancing the infrastructure needed to 'close the gap' between the information-rich and -poor (b) promote awareness of the service platforms capable of bridging this gap, and offer citizens the training needed to make use of the platforms; (c) encouraging the participation of citizens and businesses from the widest possible cross-section of the community; (d) drawing on national and European Union programs as a way of funding such developments, and (e) securing their long-term financial viability.
- Form a clear, scheduled development plan in terms of aims, priority areas and corresponding actions. Here, priority should be given to the development of small-scale, thematic or district-based projects, which would be integrated (pp. 51–52).

van den Berg and van Winden (2002) and van der Meer and van Winden (2003) detect a policy shift from the 1990s entrepreneurial approach and toward a model of local government-led actions. While many ICT-related developments in Europe during the 1990s were predominately entrepreneurial, they propose VVH and VTH infrastructures are not platforms for servicing the market but for governing them on behalf of the community. In this phase, there is instead a withering away of the market's hold over such infrastructures and emergence of an electronically enhanced government, with the potential for communities to engage with the city's infrastructures and service platforms they offer. According to van der Meer and van Winden (2003), the emerging policy framework is not a clean break from the entrepreneurial approach, but a way for local administrations to begin taking the lead in managing the relationship between what the market wants from them and the community in turn needs.

For van den Berg and van Winden (2002) and van der Meer and van Winden (2003), there are three components key to the successful management of this relationship; these are: access, infrastructures, and services. In reviewing the success of their developments in Manchester, they draw particular attention to significant achievements this administration made in delivering user-friendly access to city services and the underlying infrastructure. For unlike the top-down entrepreneurial drive of the first phase, here the infrastructures underpinning these service platforms support what is referred to as a "bottom-up transformation of communities." That is, a bottom-up transformation successful in:

- Targeting the social need of communities
- Allowing communities to form a wide range of stakeholder groups from which they can not only develop their own contributions but also learn from studies of successful ventures that are taking place elsewhere in Europe

- Setting up a single agency to oversee the city's ICT developments and manage the underlying infrastructures supporting city service platforms
- Progressively building on the capacity these infrastructures have to develop a more extensive platform of web-based services, capable of closing the gap between what the market offers and communities need in terms of employment opportunities, education, housing, and crime prevention
- Getting these infrastructures to support the delivery of such service platforms to the neighborhoods of New East Manchester
- Closing the gap between the levels of service provision within neighborhoods and across the City of Manchester by basing them on standards of equal measure
- Bridging the "digital divide" by matching the actions taken to meet the social, environmental, and economic objectives with the dividends available to sustain the development of such bottom-up transformations

### *11.3.3 Phase 3: Next-Generation Broadband*

Acting on the success of this bottom-up transformation, the City of Manchester took the decision to consolidate this governance model under the third (post-2004) phase. As they state in the Manchester Digital Strategy Progress Report (Mycio and Carter 2008).

> In January 2007 Manchester City Council submitted the ONE-Manchester Partnership Digital Challenge proposal to Government with plans for developing: "universal, affordable next generation broadband access" which "is essential to connect all residents and businesses of the Manchester City-region to the social, educational, informational and economic opportunities they deserve." (p. 5)

This established the foundation for the creation of a Manchester Digital Strategy offering

> the most advanced 'next generation' connectivity in the UK, providing a sustainable base for high growth business, innovation, transformational public services and an inclusive knowledge society. (p. 5)

The main aim of the project was to provide accessible and affordable broadband, based upon proven state-of-the-art capabilities of fiber and advanced wireless, as is currently being demonstrated across continental Europe. In cities such as Amsterdam and Paris, fiber connectivity is available to all at 100–1,000 times the speed of current UK broadband and at a fraction of the cost.

ONE-Manchester's idea of turning the digital divide into a digital dividend has been about enabling everyone in the community, no matter how excluded and disadvantaged, to gain a stake in the knowledge economy and to use this as a means to provide themselves with a better life, particularly in terms of work, skills, and health.

The first stage of ONE-Manchester's journey of transformation began by focusing on local communities. Examples from New East Manchester (Eastserve) and eTameside demonstrated the potential of community-led ICT programs, the achievements of which can be attributed to user-generated online content. The ONE-Manchester project would therefore draw upon the successes of these local projects to develop a new set of services designed both by and for the user community of citizens and businesses.

The next stage saw the development of a new tool, aimed at providing everyone with the digital technologies able to transform their lives. ONE-Manchester was, therefore, about building on the city's experiences and using the partners' proven track records in delivering projects which transform lives. This started by taking people's real experiences of both the challenges and the benefits of using technologies in this way, illustrated through a series of user journeys based on archetypes developed through community engagement and consultations.

### *11.3.4 Stakeholder Accounts of the Emerging Governance Structure*

Here, the comments of the key stakeholders involved in these ICT-related developments and affected by the emerging system of governance are reported on. These are drawn from a series of structured interviews with 30 members of Manchester's academic, industrial, and administrative communities undertaken in between December 2010 and January 2011. All questions were set in advance by IPPT, with interviews conducted by Edinburgh Napier University. Responses to the questionnaire are grouped according to the sector of the community represented. The interviews were designed to capture the experiences of those within the said communities who had been involved in the ICT-related developments, so as to solicit their views and opinions on the emerging system of governance. The stakeholders were sent the loosely structured and open-ended questionnaires prior to the interviews and asked for their comments on the questions posed. Given the strategic nature of the developments, the interviewees were encouraged to offer a "high-level" account of their experiences.

Reflective in nature, the objective of the interviews was to track what might be best referred to as the discursive elements of the ICT-related developments and to generate critical insights into the emerging governance systems. It was anticipated the insights gleamed from this exploration would not only complement the wider comparable study but also make a constructive contribution to the formation of policy on the development of such systems across Europe. How the questionnaire, loosely structured interviews, high-level accounts, discourse, and critical insights both complement the wider study and also make a constructive contribution to European policy on the development of such governance systems is reported in

Misuraca et al. (2011). What follows restricts this account of the developments and emerging governance systems to the City of Manchester.

The questionnaire is formative in nature, designed to review the developing systems from the perspectives of the key stakeholders, cross-reference their accounts, and authenticate them as part of a triangulation of the underlying discourse on ICT-related city governance. Unlike most questionnaires, it focuses on the experience of the intermediatory stakeholders responsible for developing the city governance systems rather than the user community consuming the services in question. This is because the underlying purpose of the questionnaire is to offer a discursive analysis of the developments and contribute to the ongoing review of the city governance systems emerging across Europe (EUROCITIES 2011).

In view of this, what follows does not attempt to offer a detailed account of the questionnaire, the views, opinions, and comments received from the interviews as some kind of verbatim account of the exchanges. Instead it follows the logic of the exercise by offering excerpts of the responses that manage to both draw attention to and also analyze the discourse as part of an emerging semantic on the city governance system(s). The questionnaire used for the interviews is set out in Appendix.

### *11.3.5 The Academic Community*

Representatives from the academic community cited the obvious advantages of ICT in enhancing the services Manchester offers the city. They acknowledged changes to the internal machinery of governance, such as the committee meetings, planning applications, where online document repositories, e-mail document circulation, and shared databases have led to increased efficiency and transparency. Interviewees also noted that many of the city's services now offer online information and support and commented on what they saw as the city's attempts to engage residents via ICT-enabled resources. However, all of the academic interviewees commented on digital exclusion in Manchester as a key barrier to the city using ICTs to engage the community in decision-making. One respondent commented that, due to the extent of the "digital divide" in Manchester, "online isn't going to replace paper any time soon." It was generally expressed that digital exclusion is part of a far wider problem facing the governance of Manchester, namely, the continued socioeconomic deprivation experienced by a large proportion of city residents.

All of the academic respondents commented on the "pockets of good practice" in digital skills initiatives across the city, mentioning Eastserve, EVHs, and the work of MDDA in supporting business-related IT developments. The coordination of these small projects was questioned, in that the respondents were unclear as to the city's overall strategic direction in terms of rolling out some of the piloted services to the rest of the city, with one respondent suggesting that the "profound lack of coordination" (at senior management level) was to blame.

All respondents agreed that Manchester has responded well to "the Internet revolution" and that the city has been successful in attracting new investment. Indeed, the growth of "media-based activities" was seen as pivotal to the evolution of Manchester as a postindustrial city. One respondent commented on the importance of the city's investment in the broadband infrastructure, noting that it contributes toward making the city more competitive. Communities were also highlighted as beneficiaries of new digitally enabled services, in that such organizations are able to function more efficiently using technology. However, one respondent noted that, to their knowledge, evidence suggests that community organizations only make use of basic technologies to function more coherently rather than more complex applications.

The academics did not, in general, feel that there was much evidence of enhanced citizen engagement using ICTs. One felt there is a problem with participation in general, in that it is currently difficult for citizens to become involved in local strategic partnerships, but did feel that ICTs could make a difference by enabling intelligent, informed conversation among all kinds of stakeholders. Another commented that, despite the city's attempts to tackle the digital divide, "the impact [in terms of] increasing skills and reducing unemployment are not apparent." It was suggested that the city needs to work on building residents' capacity to be involved in the management of their neighborhood rather than simply developing ICT-enabled governance services.

### *11.3.6 Industry and Local SME Sector in Particular*

Representatives from local SMEs made little comment upon the impact of ICTs on regulatory and legal frameworks, but did note that they felt further work needed to be done on this front. One interviewee felt that Manchester could use legislation to enforce sections of the city's digital plan, such as on the issue of inclusion, suggesting that any business wishing to build, for example, a wireless network in the city could be encouraged to implement something similar in another, perhaps in a less "digitally enabled" part of the city, to create a more balanced approach.

In general, the SMEs saw the British model of "siloed" governance as a barrier to the "radical" transformation they thought ICTs could offer. They agreed that cities have the potential to play a significant role in the "information society," but felt the City of Manchester to be relatively slow in developing and implementing its digital strategy. The SMEs interviewed had all been engaged in collaborative work with the city, largely in delivering social inclusion projects. All were very supportive of the MDDA and its role in driving innovation, particularly in the work it undertakes to support the city's SMEs in acquiring the skills they need to survive in the "digital age."

The interviewees recognized the successes the MDDA has achieved in pilot projects, such as Eastserve, but were keen to point out that innovative ICT projects tend

to be localized and that Manchester does not appear to be coordinating the rollout of ICT-driven services across the city. Again, this was seen to be the responsibility of senior management within the city who, the interviewees felt, ought to firmly embed the digital agenda within the city's main development strategy and to "drive change from the top."

### *11.3.7 The City's Administration*

Interviewees from the City of Manchester's administration felt that ICTs had significantly impacted upon the day-to-day running of the city administration. In particular, the city representatives highlighted the increased efficiency of the organization, in that ICTs have allowed the city to reduce its workforce over the last 20 years and automate formerly resource-intensive procedures. The new "self-service" web channels, enabling residents to interact with the city online, were cited as "quicker and more efficient and get the job done more easily," as opposed to the high cost attached to face-to-face work. One representative noted that the savings made in automating certain city services enabled the administration to use these savings in engaging with those residents who require the most support. All the city representatives felt that ICTs offer the potential for the city to make fundamental changes to service delivery, but cautioned that a large number of the city's residents are "still digitally excluded." One representative mentioned a study undertaken by the University of Manchester where approximately two thirds of the residents approached stated that they still preferred telephone or face-to-face interaction with the city, suggesting that there are issues other than lack of Internet access behind the poor take-up of online services.

In addition to ICTs enabling more efficient customer services, city representatives discussed the changes to the back-office structure and working practices of the administration. In 2009 the city appointed an acting CIO from the private sector; under their leadership, Manchester's IT Service was moved from within the Department of Finance to the Department of Transformation and the city's digital strategy was devised. This digital strategy explicitly connected the issues of back-office systems/front-office online service delivery with digital exclusion, clarifying the key role of MDDA as the driver of social and economic regeneration by way of and through the use of digital technologies. All of the interviewees mentioned the ICT problems the city has experienced in recent years, namely, the Conficker virus attack of 2008/2009 which was costly and time-consuming to rectify.

Mobile working was cited as a good example of the advantages of ICT implementation. City staff whose work is based within a specific neighborhood are now able to work within that neighborhood rather than working from a desk in the town hall. One interviewee mentioned the refurbishment of the town hall, which is currently underway, and cited it a good example of the city taking the opportunity to

modernize the working environment. Staff have been moved into a new building for the duration of the refurbishment, and this new building is designed for a mobile workforce: it is open-plan and all desk space is available for any staff member to use. The city's ICT setup allows any staff member to log in to any of the PCs, and, crucially, there are fewer desk spaces than there are staff "on the basis that most people are out-and-about delivering services." Staff working outside of the office use mobile phone and smartphone technologies to connect to the documents they require. The promotion of mobile working has resulted in significant savings, and the newly refurbished town hall will be designed and equipped to support this modernized working environment.

The advantages ICT-enabled services bring to citizens are largely related to their ability to access information online. The interviewees felt that the city's role is "to keep people informed … ensuring that our citizens have access to information." One respondent mentioned that key committee documents are now posted online, such as agendas and minutes, thus opening up the decision-making process to greater public scrutiny. Another mentioned that some city services now make use of social networking tools to keep in touch with customers and to get feedback on services, citing the Library Service and their use of Twitter and Facebook as an example. All interviewees mentioned the Leader's Blog as an example of the city harnessing social networking to open up a "direct communication link" to the top of the council. Through his blog, the Leader is open to questions and criticisms, which city representatives felt contributes toward the increased transparency of the organization. There does not seem to be much interest in using ICTs to enhance public consultations. Representatives mentioned that pilots had very limited take-up and that, "rightly or wrongly, politicians regard [this] as tokenistic." Interviewees felt that the city has strong governance structures in place for public consultation and that it was more important to find the most appropriate ways of engaging with people, in a wider sense, than simply looking at ICT-enabled options.

## 11.4 Focus on the Next Generation Digital City

Acting on the success of this policy framework, the City of Manchester took the decision in 2007 to consolidate their governance model. They stated, in Mycio and Carter (2008), that

> In January 2007 Manchester City Council submitted the ONE-Manchester Partnership Digital Challenge proposal to Government with plans for developing: "universal, affordable next generation broadband access" which "is essential to connect all residents and businesses of the Manchester City-region to the social, educational, informational and economic opportunities they deserve." (p. 5)

ONE-Manchester did not win the Digital Challenge competition,[2] instead taking the "highly commended" runner-up place, but preparation for the bid included the creation of a Manchester Digital Strategy with a vision of creating the city region as

> the most advanced 'next generation' connectivity in the UK, providing a sustainable base for high growth business, innovation, transformational public services and an inclusive knowledge society. (p. 5)

The digital strategy and its proposals for a "Next Generation Digital City" (Mycio and Carter 2008) aim to provide accessible and affordable broadband which is based on the proven state-of-the-art capabilities of fiber and advanced wireless, as is currently being rolled out across continental Europe. The strategy continues to underline the "ONE-Manchester" notion of "turning the digital divide into a digital dividend," enabling everyone in the community, no matter how excluded and disadvantaged, to gain a stake in the knowledge economy and to use it to provide themselves with a better life, particularly in terms of work, skills, and health (Carter 2009).

The Digital City proposal is aimed at transforming Manchester into a "world-class exemplar" of how digital technologies can be used to support economic growth, continue the transformation of public services, tackle the digital divide, and create "inclusive sustainability." The key driver behind the project is investment in next-generation Internet connectivity, namely, fiber to the premises (FTTP) and advanced wireless. These technologies, particularly FTTP, are currently utilized by cities across Europe, providing the city with a number of case studies in the use of the technology and emerging business models: knowledge and experience the city intends to harness in its analysis of the technological options available.

The creation of "digitally inclusive sustainable communities" remains at the core of the Digital City project. The digital strategy (Mycio and Carter 2008) states that the project

> … includes enhancing the 'sense of place' with digital technologies, being proactive in determining what a particular place, both individual neighborhoods and the city as a whole, needs and what kind of infrastructure and investment is required to meet those needs. We want to ensure that next generation connectivity is, on the one hand as advanced and future-proofed as possible based on the 'open network' principle, and on the other as accessible, affordable and inclusive as possible. (p. 7)

The city's initial objective is to develop the new high-speed Internet network in phases starting in the Oxford Road Corridor and its wider area of benefit, the Central Manchester Regeneration Partnership area. This will deliver next-generation connectivity to residents, businesses, and institutions to support job creation, skill development, business growth, and transform service provision. Further development to expand the network, starting with East Manchester, including the creation of a major Internet Hub, supporting a creative industries cluster at Central Park, will be planned concurrently with the Oxford Road Corridor program in order to maximize local benefit.

[2] Sponsored by the national government's Department for Communities and Local Government.

The city continues to work at national level within the DC10plus network, founded from the ten local authorities involved in the 2007 Digital Challenge competition. The network's vision is to promote technology and innovation as an agent for empowering people and connecting communities. Manchester coordinates the network's next-generation connectivity work-stream, focusing on developing a Living Lab model as a test bed for the development of a "Broadband Atlas" (ONE-Manchester 2007). The Atlas represents an effort to map out broadband supply issues and to stimulate feedback regarding the needs and desires of a broad user community.

### *11.4.1 Activities and Results*

The digital strategy also sets out to enhance the achievements made in the transformation of the city's physical infrastructure over the past decade. This includes adding value to achievements made with inward investment, including MediaCityUK, but also allowing for very rapid expansion for other development sites within Manchester itself, especially in key regeneration areas such as East Manchester. In addition the Next Generation Digital City proposals focus on ways that digital inclusion would support the development of sustainable communities, including new intelligent "eco-solutions" that would have a positive impact in areas such as energy management, more sustainable mobility, teleworking, and telecare applications.

The delivery of next-generation fiber broadband to the regeneration program is of fundamental importance. The rollout is phased, with initial installation to 1,500 homes and businesses in 2010. The immediate impact for the resident and business community is speed of Internet connectivity: the fiber-optic infrastructure promises a minimum of 100 MB per second for both downloading and uploading. The services are available from different providers—in developing an open-access network, this digital infrastructure known as "Corridor Manchester" enables all service providers to lease the optical fiber to provide services to their customers, including TV, telephone, and other data services. In addition to paid services, Corridor Manchester anticipates that some services and applications can be free and intend to deliver innovative public services over the network, such as telecare which would allow patients to connect to their healthcare provider via video link, allowing patients to monitor their health and notify results via the phone line (Corridor Manchester 2010a, b).

The infrastructure itself is intended to encourage innovation by allowing any user to experiment and test content, in a manner likened to income-generating iPhone apps. MDDA currently offers free events on using digital technologies for the voluntary and community sectors.

### *11.4.2 Impacts on City Governance Developments*

From the interviews undertaken, urban regeneration was cited by a number of interviewees in terms of exemplifying Manchester's strong partnerships with the private sector. One city representative spoke of the city as a hub of intelligence and information and that the city's approach is to view everyone as a stakeholder, with the duty to work together "for the good of the whole city." An interviewee from the business sector in Manchester echoed this, reporting on their own involvement in local collaborative projects. These include delivering "digital inclusion projects" in the more deprived neighborhoods, such as social media workshops.

Representatives from the City of Manchester reiterated their commitment to the development of a strong working relationship between the business sector, university, and community organizations. One example provided of this was the Academies program, where the city recognized the skills and experiences of the private sector in delivering education and sought their influence in the design of the Manchester Academies model. The transformation of secondary education marks a strategic response to the pattern of low educational attainment and lack of skills across Manchester's population and reflects the city's efforts to prepare young people for the "knowledge-intensive" employment opportunities available within the city center.

Five new academies have now been built as a result of the £146 m investment. Each academy is sponsored by an industrial partner, from sectors the city describes as, "reflecting future growth." One of the representatives from the private sector (an SME) also highlighted the new Academies model as being "progressive" in terms of the investment in, and deployment of, new technologies. These include a system whereby parents can monitor their child's progress and a cashless biometric thumb print scanner for school lunch payments, in addition to investment in mobile computing technology such as palmtops, pocket PCs, and netbooks. The Academies model is to be rolled out across all the city's secondary schools, with each extending its opening hours enabling community-wide access to the facilities on offer. The key objective in opening the newly refurbished schools to the parents and wider community is to reach people who may not have access to, or knowledge of, ICTs and to enable them to use online city services.

The city's commitment to partnership with its neighboring local authorities is now formalized: the unitary area of Greater Manchester, also known as the Manchester City Region (coordinated through the Association of Greater Manchester Authorities), became a formal administered combined authority in April 2011. Essentially, this constitutes a devolution of powers from the national government to the new authority, including transport, housing, skills, and regeneration. One city representative explained that working as a combined authority, the local authorities could pool their resources "for the common good": pilot projects are currently underway in neighborhoods across Greater Manchester, trialing new methods of delivering public services to families experiencing worklessness.

## 11.5 Key Findings

In line with the recommendations made by Fourkas (2002), the governance model adopted by the City of Manchester has sought to:

Build upon the success of the 2007 Digital Strategy by providing accessible and affordable broadband, based on the proven state-of-the-art capabilities currently being rolled out across Amsterdam and Paris.

The Digital Strategy continues to underline the ONE-Manchester notion of turning the digital divide into a digital dividend; enabling everyone in the community, no matter how excluded and disadvantaged, to gain a stake in the knowledge economy and use it as a means to provide themselves with a better life, particularly in terms of work, skills and health.

The recent Digital City proposal is aimed at transforming Manchester into a "world-class exemplar" of digital technologies supporting economic growth, transforming public services, tackling the digital divide and creating "inclusive sustainability". The key driver behind the project is investment in next-generation internet connectivity, namely fiber-to-the-premises (FTTP) and advanced wireless. These technologies, particularly FTTP, are currently utilized by cities across Europe, thereby providing the City with a number of case studies in the use of the technology and emerging business models; knowledge and experience it intends to harness in its analysis of the technological options available.

The strategy adopted further enhances the achievements made in the transformation of the city's physical infrastructure. This includes adding value to achievements made with inward investment, including MediaCityUK, but also allowing for very rapid expansion for other development sites within Manchester itself, especially in key regeneration areas such as East Manchester. In addition to this, the strategy also focuses attention on how digital inclusion can support the development of sustainable communities, including new intelligent 'eco-solutions' themed around: energy management, mobility, teleworking and telecare applications.

These developments are of particular interest for the reason they signal a policy shift from the entrepreneurial approaches driving developments embarked upon during the 1990s and towards a model of local government-led actions. For with the latest phase of development (2004 onwards) the City begins to see the emergence of infrastructures not as platforms servicing the market but for governing them on behalf of the community. Here, the market's hold over the development starts to wither away as part of the transition to an electronically-enhanced governance model with the capacity to coordinate, steer and assemble the means needed for communities to access such infrastructures and platform of services they offer. The City of Manchester's experience of these developments suggests there are three components key to their success. These are: access, infrastructures and services and, in particular, user-friendly access to the underlying infrastructures and platform of services.

These bottom-up developments have been successful in targeting the specific social requirements of communities. Community stakeholders are invited into partnerships whereby they can coordinate and steer delivery of the available services

and at the same time learn from similar case studies from across Europe. The success of these community-level interventions has led to the development of a more extensive platform of web services, representing an attempt to close the gap between what the market offers and what the community needs, particularly in terms of employment opportunities, education, housing, and crime prevention. Efforts were made to close the gap between the levels of service provision both within neighborhoods and across the city, by basing them on the same set of standards.

While these developments have been embodied in the digital strategy's proposal to turn Manchester into "digitally inclusive sustainable communities," it is evident that changes to the governance system and anticipated socioeconomic changes they are having are not universally accepted across the stakeholder community. Representatives from the academic community suggested there was too little evidence to suggest that the city's digital inclusion agenda has had any real impact upon socioeconomic deprivation, or community empowerment, with one interviewee commenting

> My own feeling is that most technologies reproduce and perhaps even magnify the kinds of inequalities that already exist in society, and the transformative power of ICTs … is not empirically obvious, and I would be skeptical.

Another commented that

> … despite all the specific policies the actual impact, in terms of increasing skills and reducing unemployment, is not apparent.

The academics expressed concern at the enduring socioeconomic deprivation in the city, referring to the "doughnut of decay" surrounding the (largely regenerated) city center. It was suggested that

> … ICTs can become a palliative and the real problem is poverty … palliatives may have some effect but they don't get to the root of the problem and they certainly don't "solve" the problems that they are touted as being part of the solution for.

Some representatives from the business community remained skeptical of the city's commitment to its digital strategy. One interviewee felt that the city viewed technology as a "threat" and that "they will only [use ICT] to improve public services as long as it doesn't impinge on their own power and authority," going on to make the point that

> … by engaging with ICT infrastructure what you're actually saying is "we're prepared to govern in a different way". I don't think we have a civic leadership that is prepared to accept that.

Another interviewee from the business community envisaged a more transactional model of governance, commenting

> I'd like to see much more information made available so people can start to make decisions about how their communities are run and organized. If they succeed, maybe the Council will trade something.

Manchester's deployment of ICTs has met with considerable success. The ICT-related governance of the city is in many respects exemplary. Over the past 15 years, the city has pioneered the shift from an entrepreneurial model of ICT-related governance to a community-based solution able to support Manchester's high-profile urban regeneration strategy. Targeting the city's social, environmental, and economic aspirations, the growth, competitiveness, and cohesive qualities of this community-based governance model are at the forefront of the drive toward digital inclusion.

MDDA has been key in developing the platforms servicing this ICT-related governance model and innovative features this brings to the city. Nowhere is this currently better exemplified than in the Next Generation Digital City Strategy adopted by the administration to transform the city into a "world-class" exemplar of digital technology with the capacities and physical infrastructure developments needed to meet Manchester's social, environmental, and economic aspirations. Examples of this transformation are found in the development of New East Manchester and MediaCityUK and the Oxford Road Corridor, now forming part of Central Manchester's Regeneration Partnership.

## 11.6 Conclusions

Given the highly innovative nature of these infrastructural developments, research clearly needs to continue and provide the means to support the significant foresight developments of the kind outlined in this chapter exemplify.

Cutting across all aspects of infrastructure development, the research challenges they posses are notable. Encompassing fiber broadband rollout, platform development and service provision, shall require a multitude of hardware developments, supported by middleware service platforms and software supporting their delivery. It is therefore important to properly theme the research effort and package it so the technical and managerial expertise of the city can be harnessed alongside the major consultancies in the field. This shall require a further consolidation of the efforts made to draw funding from national and international sources and mainstream them into the EU's IST programs in particular.

From the interviews conducted, the stakeholders question how best to integrate these developments into the city's own service platforms. For while many of the interviewees noted the considerable progress Manchester is making to establish itself globally as a Digital City, the existing rollout and distribution of the technologies across the administration is patchy. This suggests there is a need to redress the current balance between the more outward-facing components of the Next Generation Digital City Strategy and use them as a means to support the integration of the administration's public service into this platform. For while some examples of how this is to be done with the likes of healthcare are available, many of the

representatives interviewed were of the opinion that this integration should be mainstreamed across all areas of service provision and therefore include health, education, and housing.

This suggests the challenge which the legacy system poses for Manchester's Next Generation Digital City Strategy should be prioritized as a research challenge, so that claims about their digital inclusivity can be substantiated, not so much in terms of leading edge innovation, but in their diffusion and systematic integration into the emerging infrastructure.

Some of these research challenges include:

- Back-office integration of services, provided across the administration by way of and through a platform able to augment existing provision in line with the technical standards and semantic definitions emerging to support the next-generation fiber broadband infrastructure.
- Use of the said infrastructure to bundle the city's service provision around recognized themes, themselves related to user needs.
- Augmentation of these service bundles via appropriate middleware supporting multichanneled access and packaged to users situated at the front end.
- The development of the city portal as a "single destination point" for service provision across the public domain.
- Use of the portal to develop the city's current level of service maturity, focusing on the augmentation of provision from the level 3 (transactional) to 4 and 5 (the transformational-government stage of development). Those ICT-related developments needed for Manchester to materialize the efficiency gains any mainstreaming of such e-democracy-based service models offer administrations.
- The formation of an action plan for rolling out this digitization of the city's services and diffusion of the technology to support this.
- The need to baseline these developments and use such profiles of service provision as a means to monitor and evaluate what regeneration actions, like those exemplified by the Oxford Road Corridor, contribute to the city.
- The selection of indicators able to measure the performance of such developments in social, environmental, and economic terms and capable of assessing what they contribute to growth, competitiveness, and cohesion.
- The types of partnership agreements best suited to manage such developments across the city and knowledge infrastructures able to sustain the improvements they offer Manchester.

From the strength of the comments received on the experiences of Manchester as a Digital City, it would seem appropriate to apply the innovations embodied in these developments to create a policy commitment that is shared across the administration. In particular, that is shared in terms of an agreed development plan, program of work, set of tasks, and resource base which can be made available to fund their implementation.

## Appendix: Interview Questions

### *Section 1: General*

1. What do you consider to be the main challenges those responsible for governing Manchester face?
2. What are the main social and economic challenges Manchester currently faces?
3. How is the city responding to them?
4. What particular action is Manchester taking? Please give examples.
5. What is your organization doing to meet these challenges?
6. What guidelines do you follow in taking in acting on these challenges?
7. What are the aims and objectives of any such actions?
8. What main tasks are you currently involved with?
9. How will completing these tasks change the perception of the Manchester?
10. How shall they improve the image of the city?
11. How will these improvements promote a positive image of the city?
12. How does all of this lead to the perception of Manchester as a "well-governed" city?

### *Section 2: ICT Developments Within the City*

1. What role can cities play in "information society"?
2. How can ICTs enhance life in the city? In Manchester in particular?
3. Have there been any changes to the governance structure of the city (resulting from ICT developments)? If so, how have these changes affected:

   (a) Businesses
   (b) Citizens
   (c) Community organizations

4. [If there haven't been any noticeable changes to date, what changes can you predict, and how might they affect the above groups?]
5. How have these changes affected the quality of service delivery, and what evidence is there to support this (if available)?
6. Have these changes affected residents' satisfaction in service delivery, and what evidence is there to support this (if available)?

   (a) Can you give an example of any opportunities and challenges that can be identified as a consequence?
   (b) What should the city do to respond to these?
   (c) What is it currently doing?

7. Does the city have an ICT action strategy? Can you provide any further information on the development of this?
8. How is ICT used within the city administration?
   (a) How has this evolved?
9. What is your opinion of the current ICT setup within the administration of the city?
10. Has there been a change to the decision-making process within the city? If so, can you provide an example?
    (a) How has this affected your work?
11. How do you keep yourself informed, with regard to the impact of ICTs on the administration of the city?
12. Which topics do you feel well informed on, and which would you like to know more about?
13. How has the city/your department worked to train employees on new ICT developments?
    (a) Do you have any feedback on this?

## *Section 3: ICT-Enabled Changes in the City Governance System*

1. What changes have ICTs contributed to the current governance model in Manchester and in what respect have these changes been made apparent? Please give reference to specific procedures and tasks that have been affected by such changes.
2. Would you expect subsequent ICT systems to further impact upon governance processes in Manchester?
3. In the last 5 years, what have been the main ICT developments that have led to the current model of governance used in the administration of Manchester today?
4. With respect to the impact that ICTs have had on the administration of government processes, which individuals/groups are driving this change and making the key decisions?
5. In your own role, what capacity do you have to influence the uptake of ICTs and subsequently shape the governance model in Manchester?
6. In making such decisions, which other individuals and groups are consulted?
7. What would you say are the main problems experienced by the current model of governance used in the administration of Manchester today?
8. With reference to ICT developments, please outline the extent to which citizens currently participate in the current model of governance in Manchester today?
9. What could be done to improve such citizen participation?

10. With reference to the current model of governance in Manchester, have there been any notable developments that have been primarily driven by the emergence of ICT?
11. What do you think are the opportunities and challenges that exist for the further application of ICT-enabled governance in Manchester?
12. How do you see yourself participating in the future ICT-enabled model of governance in Manchester?
13. In the short term, what improvements to the current model of governance in Manchester could be initiated (primarily) with the support of ICTs?

### *Section 4: Socioeconomic Implications*

1. How is the city's use of ICTs helping to regenerate Manchester?
2. What are the main drivers behind this ICT-related regeneration?
3. What do you think is making such a regeneration of the city possible?
4. What do you consider to be any possible barriers to Manchester's regeneration?
5. What do you consider to be the main risks in the actions the city is taking to regenerate Manchester?
6. How is the city making sure the regeneration of Manchester is well governed?
7. Do you think what Manchester is doing can be said to be a model of "good governance" and other cities would be wise to follow?

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